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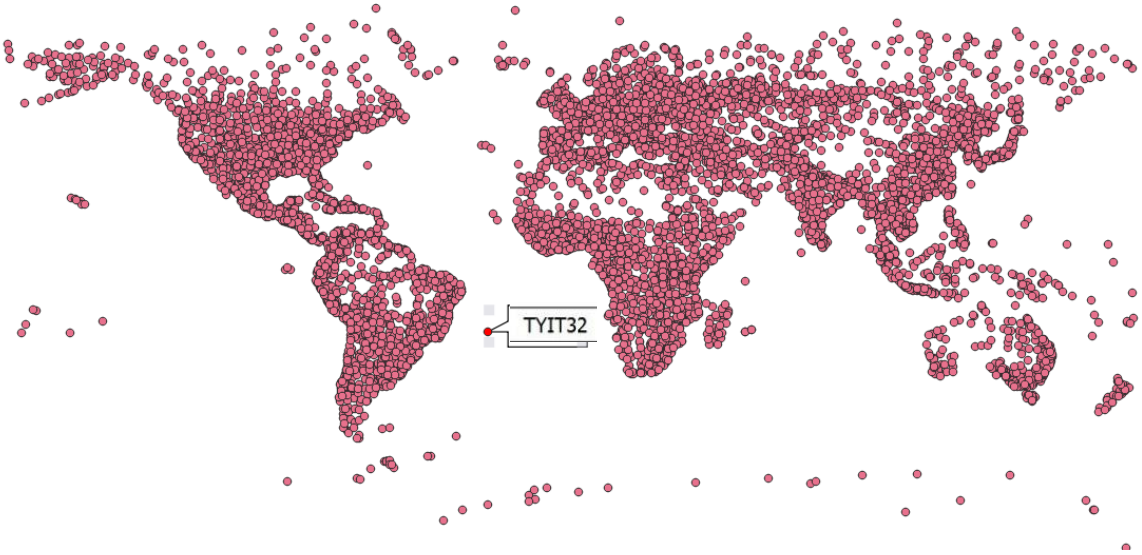
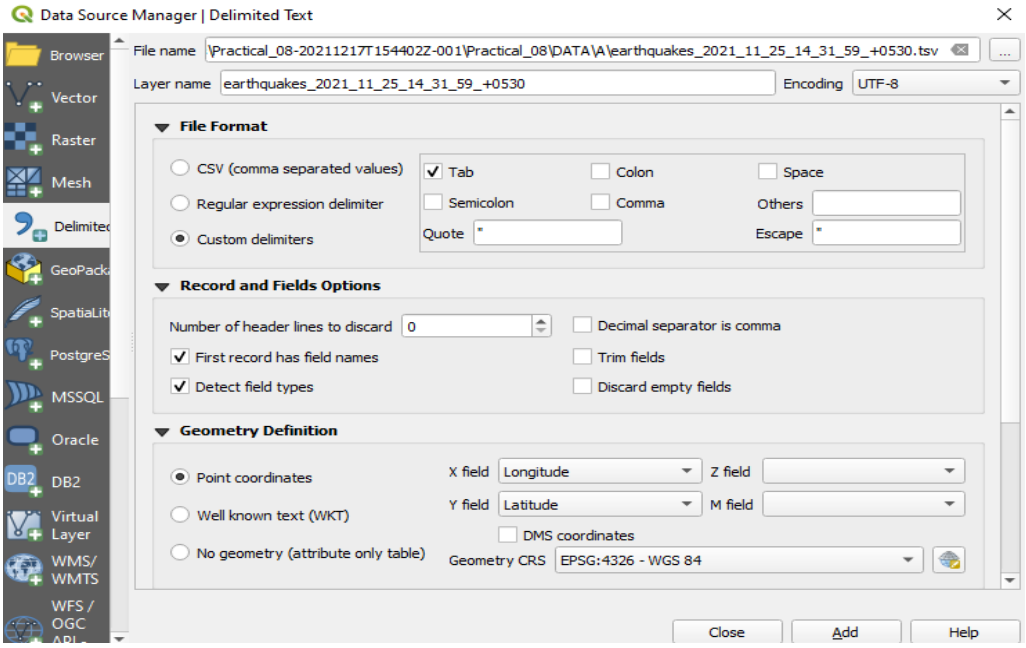
Q. Advanced GIS Operations.

A. Nearest neighbor analysis.

1. Open QGIS → new project.
2. Add Vector Layer → select file Practical_08 → DATA → A → ne_10m_populated_places_simple → ne_10m_populated_places_simple.shp → Add → close.
3.



4. Add delimited text layer → select file Practical_08 → DATA → A → earthquakes_2021_11_25_14_31_59_+0530.tsv → do changes → add → close.



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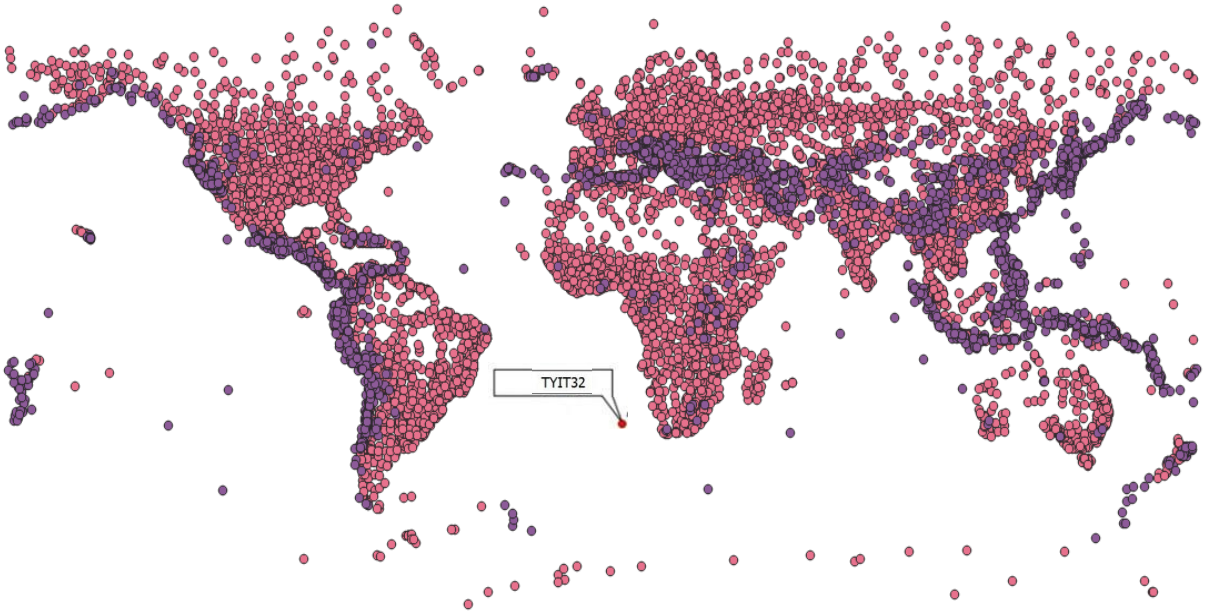
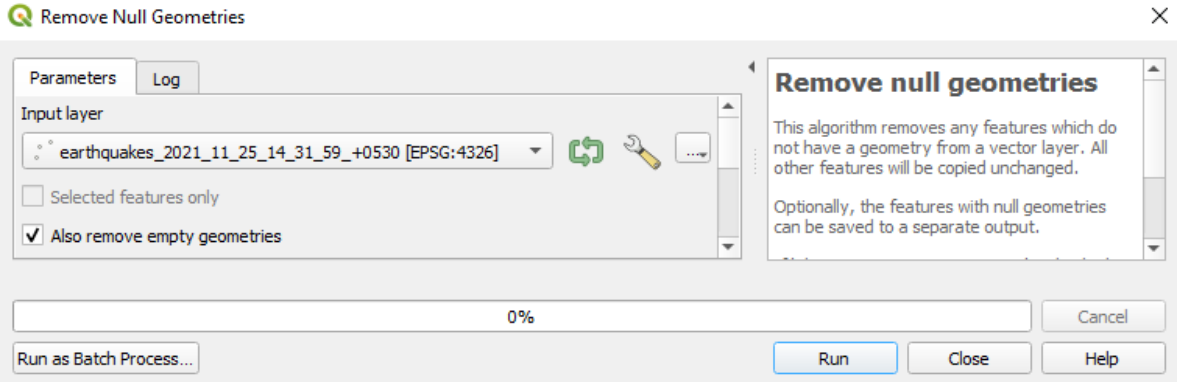
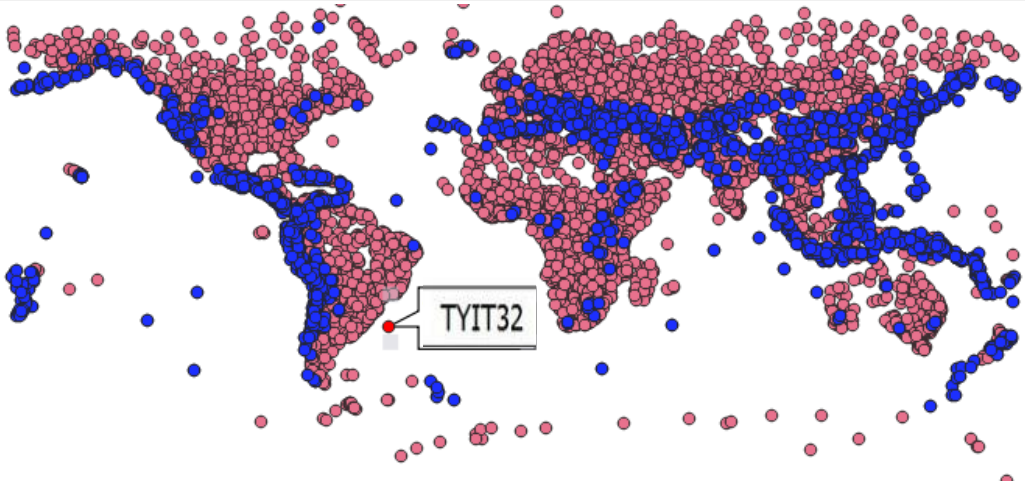


5. Click on processing → toolbox → Vector geometry → double click on Remove null geometries → do changes → run.





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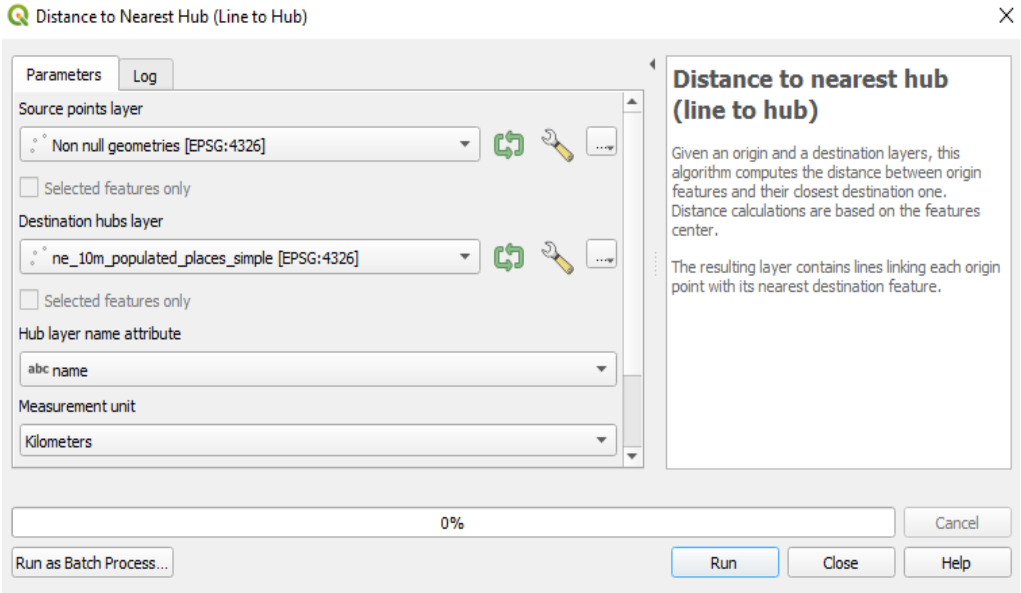
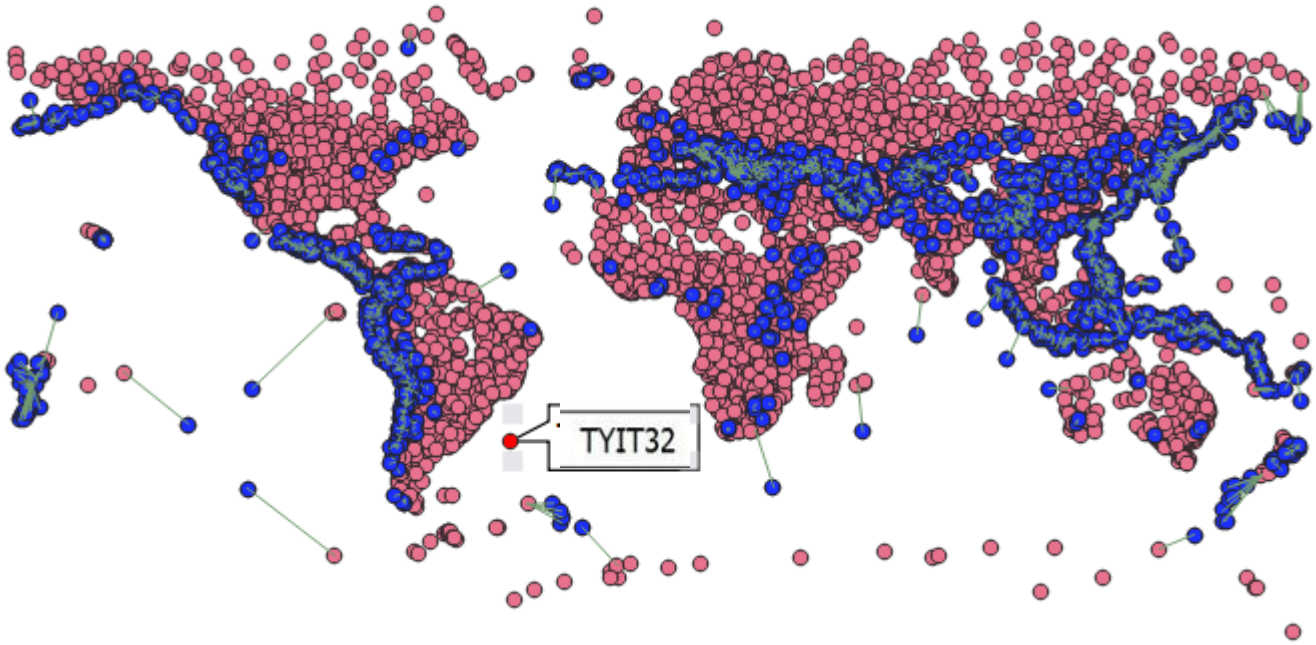
6. Again click on processing → toolbox → vector analysis → double click on distance to nearest hub (line to hub) → run.



7. Output:



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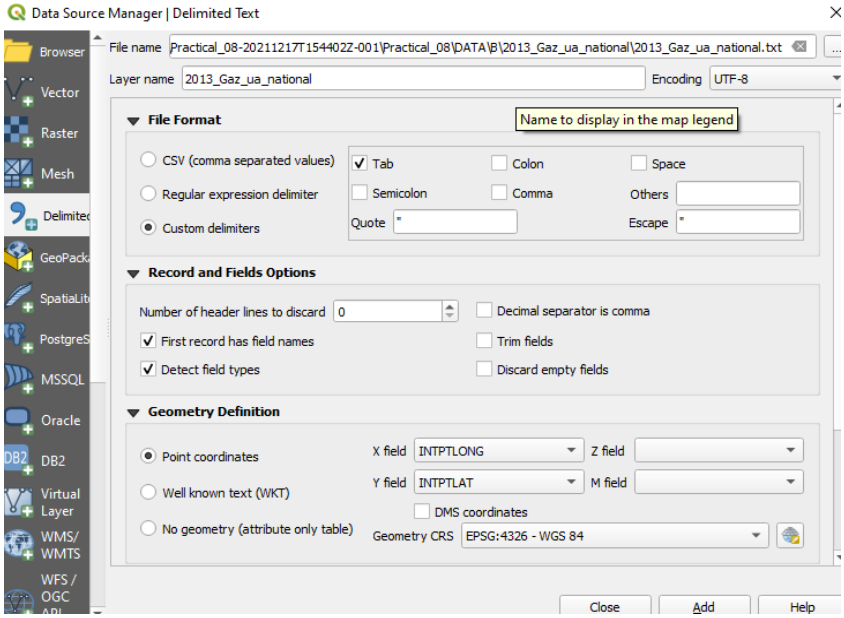
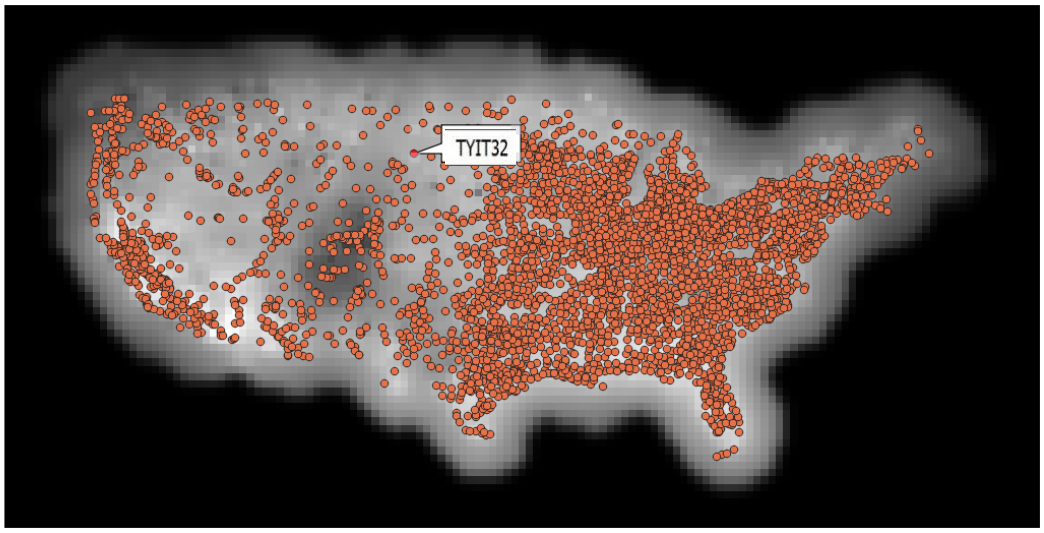
B. Sampling raster data using point or polygons.

1. New project.
2. Click on Add raster layer → select file practical_8 → DATA → B → us.tmax_nohads_ll_20140525_float.tif → add → close.
3. Add delimited text layer → select file practical_8 → DATA → B → 2013_Gaz_ua_national → 2013_Gaz_ua_national.txt → and do some changes → add → close.





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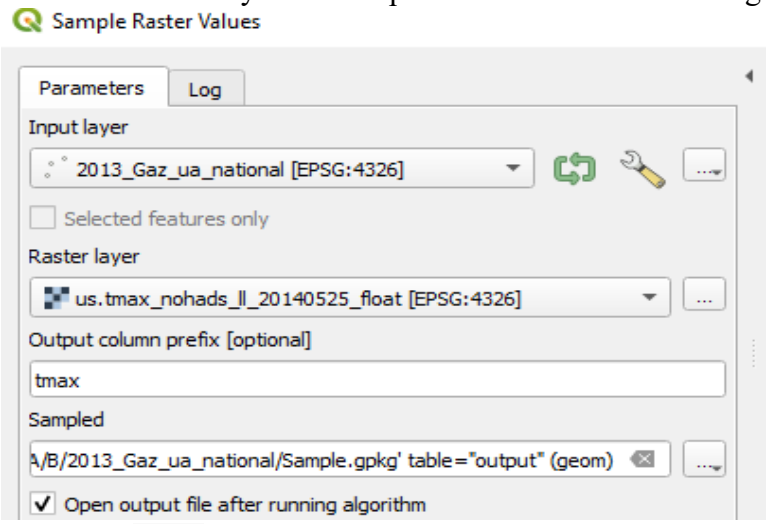
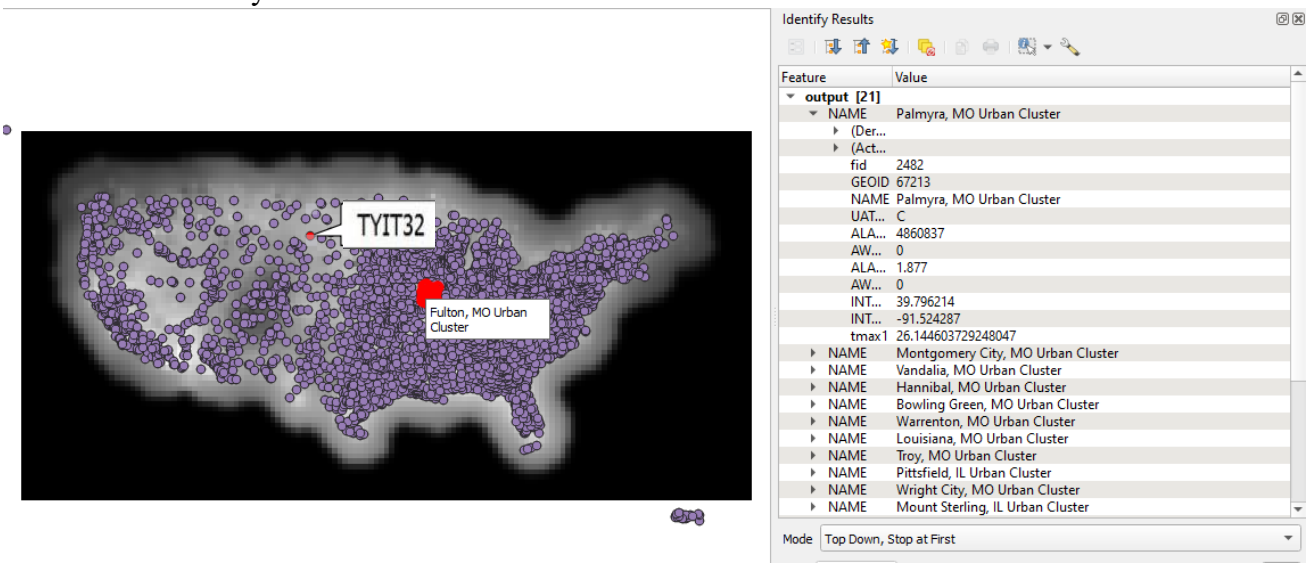
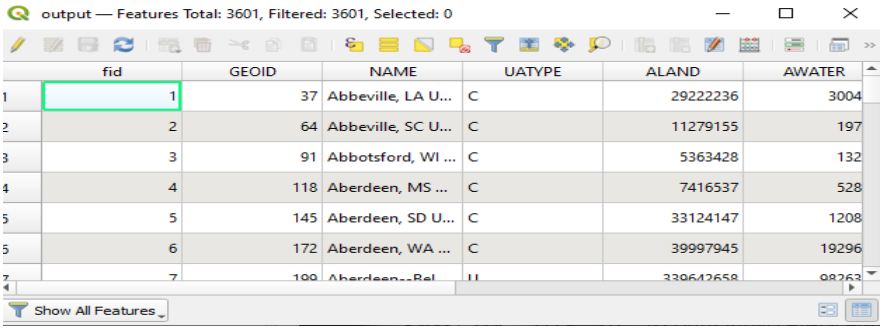
4. Processing tab → toolbox → raster analysis → sample raster values → do changes → run → close.



5. Then click on Identify features .



6. Right click on output layer → open attribute table.



	fid	GEOID	NAME	UATYPE	ALAND	AWATER
1	1	37	Abbeville, LA U...	C	29222236	3004
2	2	64	Abbeville, SC U...	C	11279155	197
3	3	91	Abbotsford, WI ...	C	5363428	132
4	4	118	Aberdeen, MS ...	C	7416537	528
5	5	145	Aberdeen, SD U...	C	33124147	1208
6	6	172	Aberdeen, WA ...	C	39997945	19296
7	7	199	Aberdeen--Bel...	U	339642658	98263

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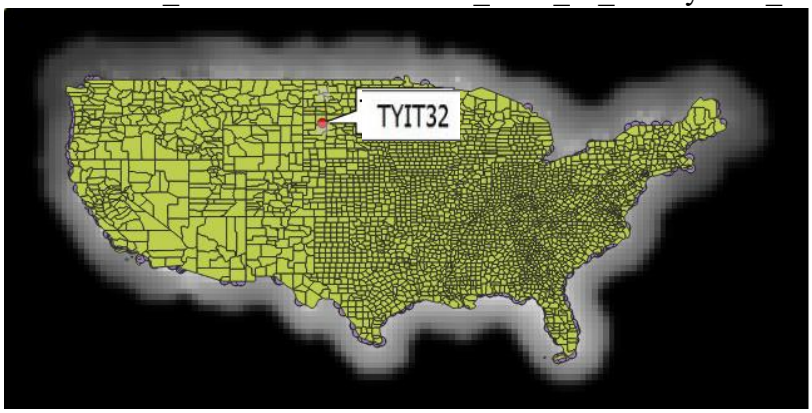
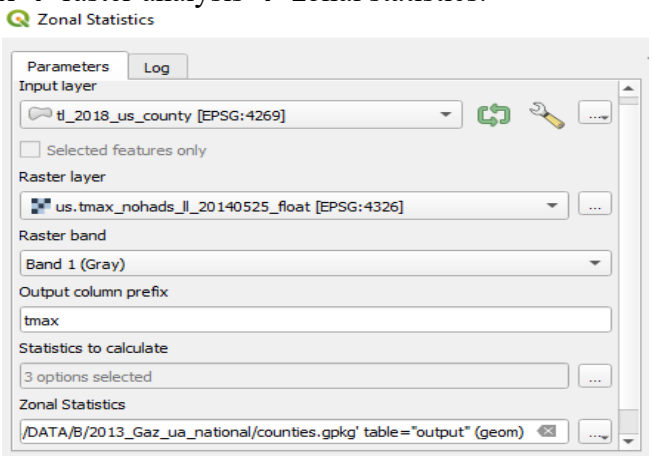
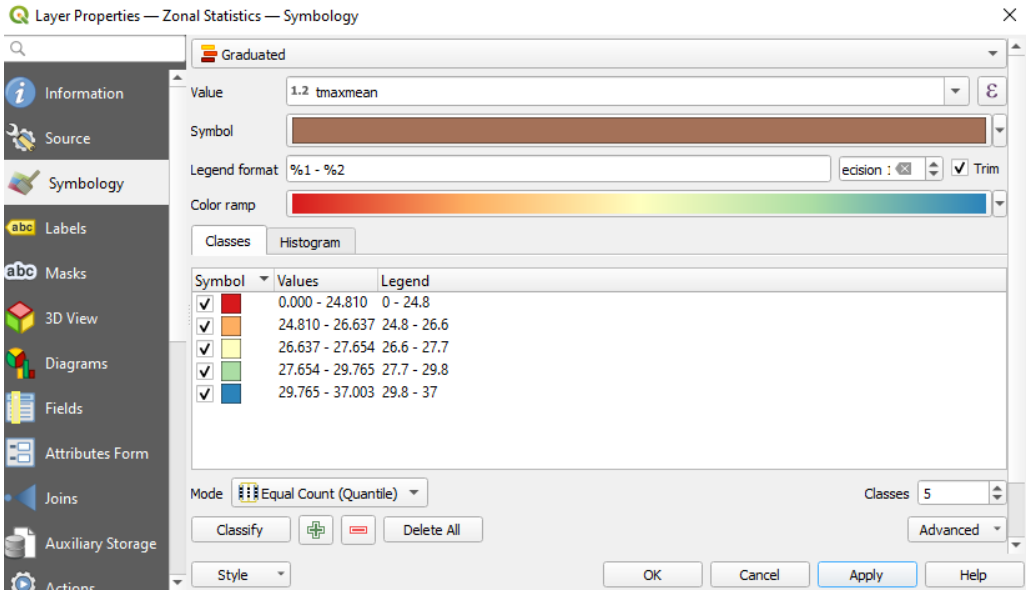
7. Add vector layer → Practical_08 → DATA → B → tl_2018_us_county → tl_2018_us_county.shp → add.



8. Processing → toolbox → raster analysis → zonal statistics.



9. Right click on output 2 → properties → symbology → do changes → apply → ok.



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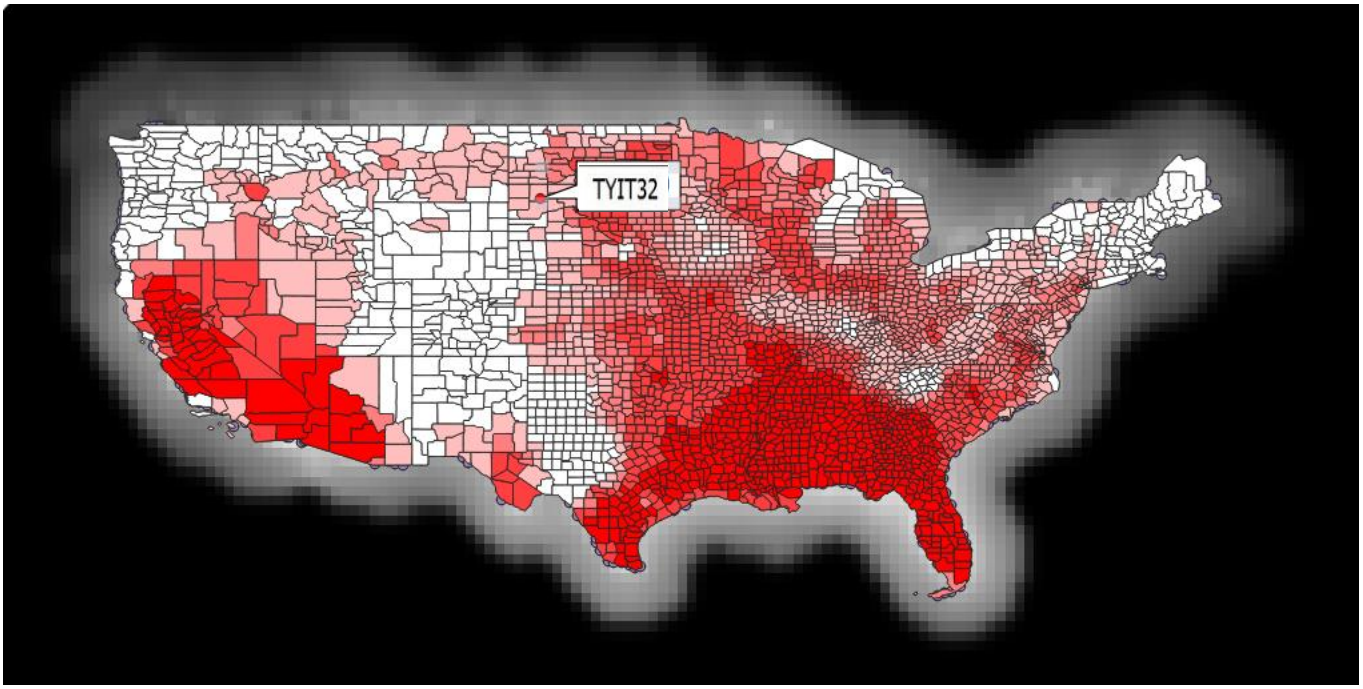
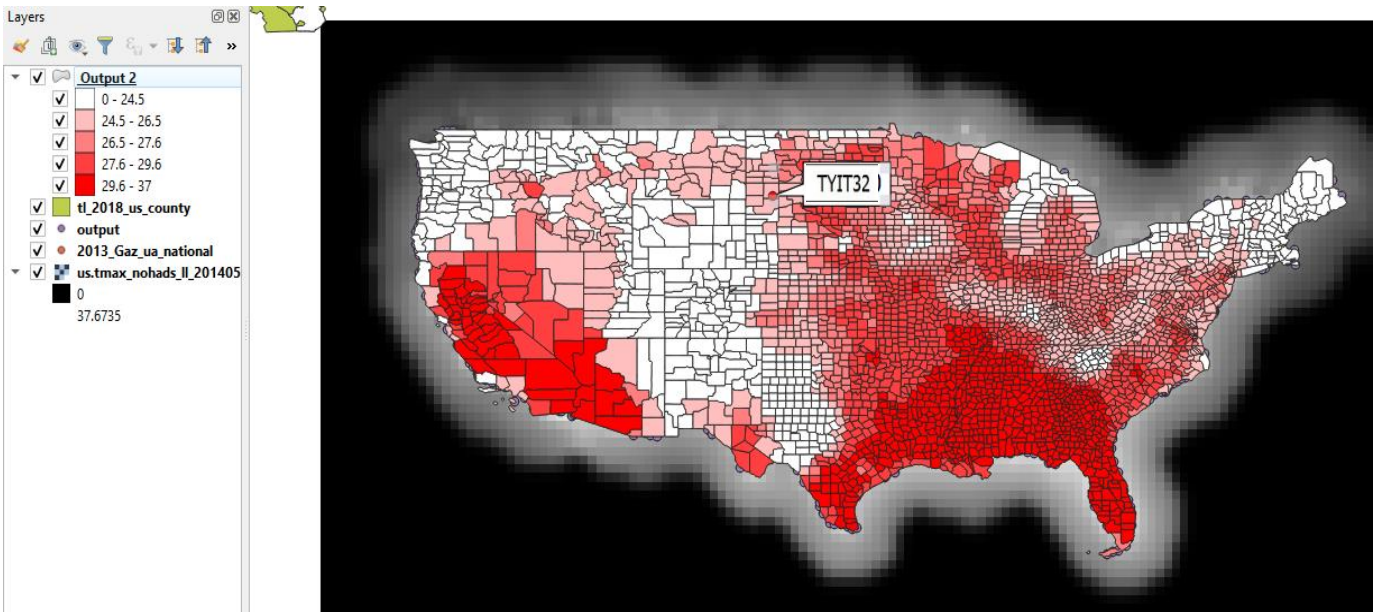
10. Output:





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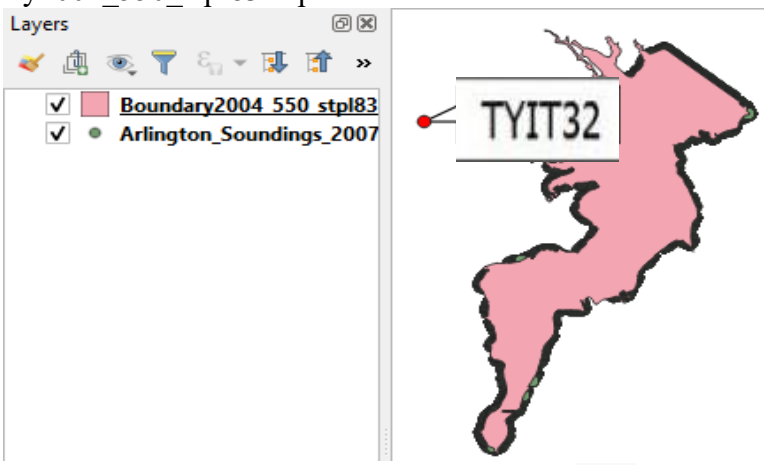
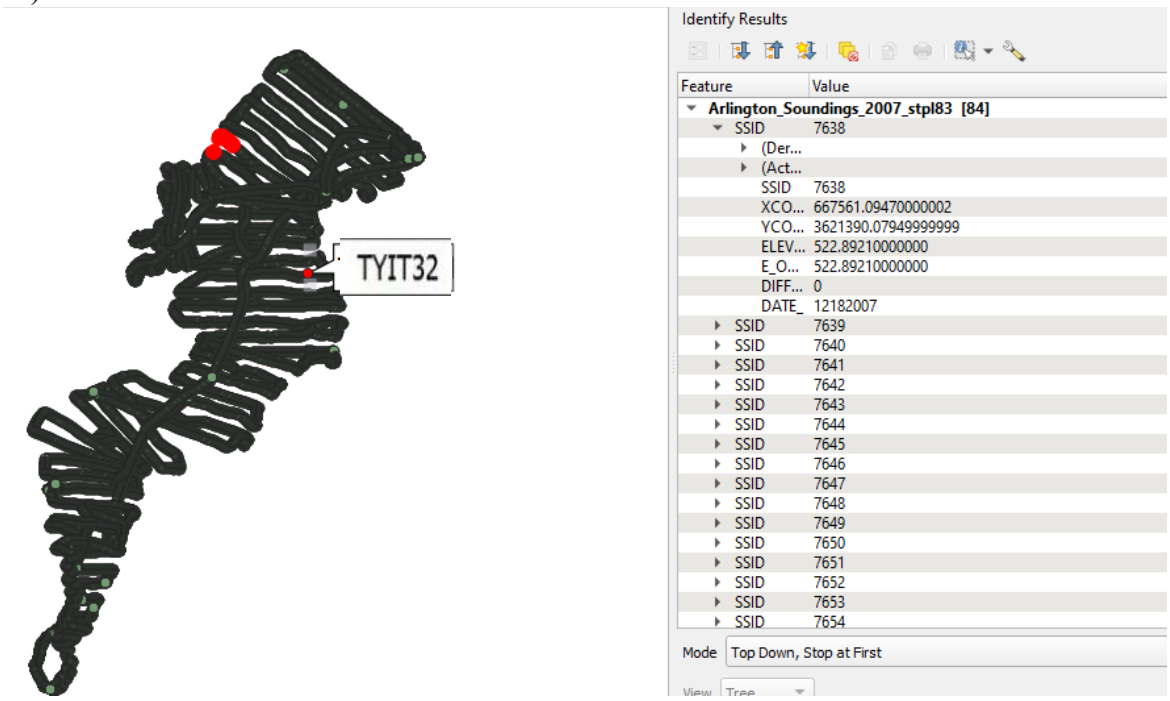
C. interpolating point data.

1. New project.
2. Add vector layer → select file practical_8 → C → Shapefiles → Arlington_Soundings_2007_stpl83.shp → add → again select file → Practical_08-20211217T154402Z-001 → Practical_08 → DATA → C → Shapefiles → Boundary2004_550_stpl83.shp → add → close.



3. Unselect Boundary2004_550_stpl83 → click on identify features → right click on dot → identify all(1331).



4. Add vector layer → Practical_08 → DATA → C → Shapefiles → Islands_2004_550_stpl83.shp → add → close.

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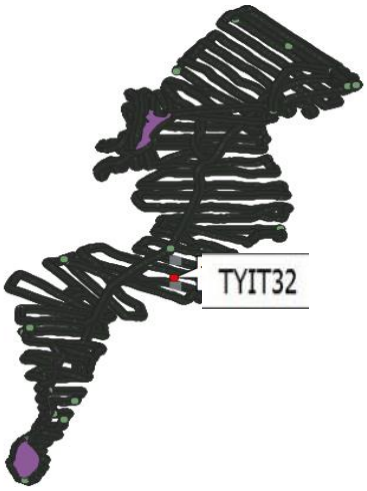
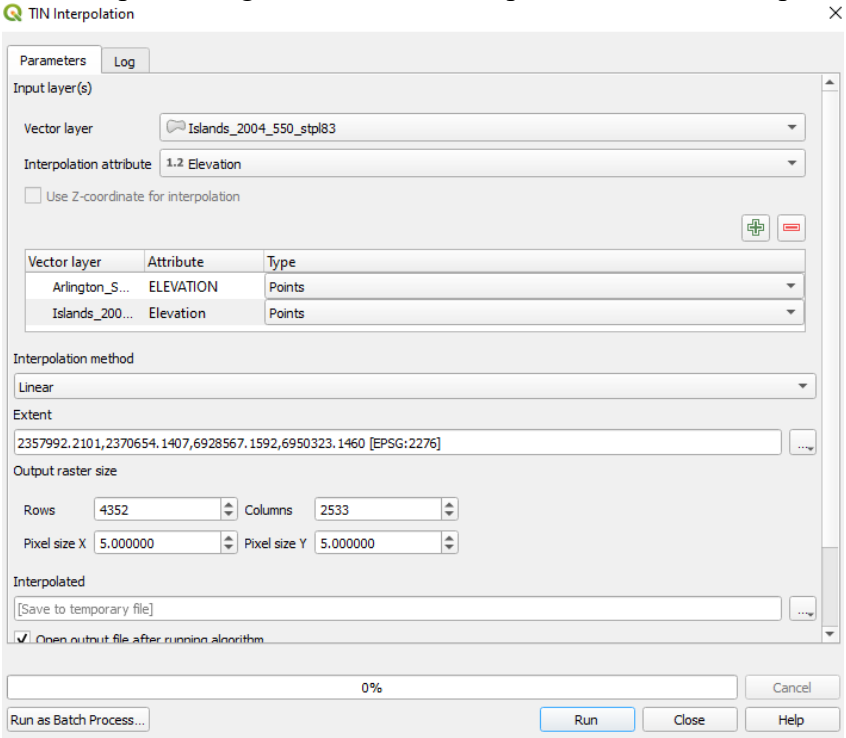
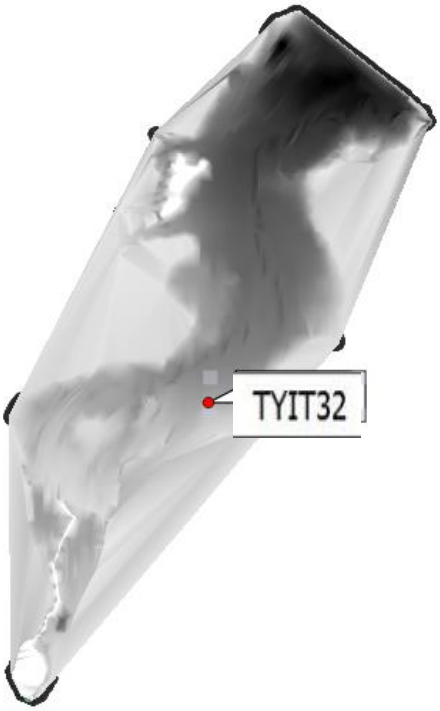
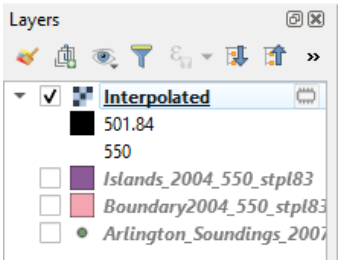


5. Click on processing → toolbox → interpolation → TIN interpolation → do changes → run → close.





6. Accept Interpolated other unselect.



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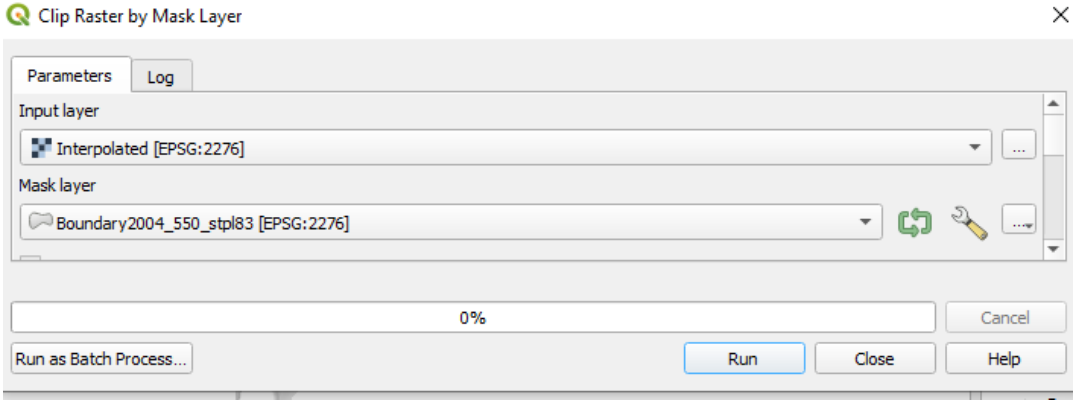
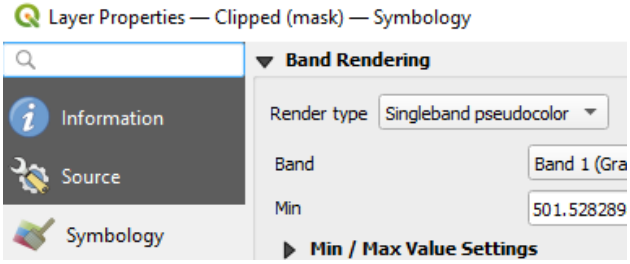
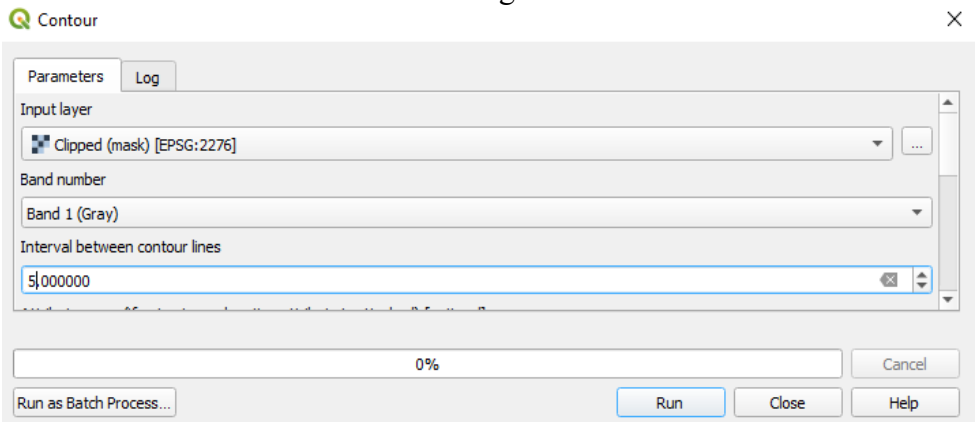
7. Processing → toolbox → GDAL → raster extraction → clip raster mask layer → run → close.(unselect other)



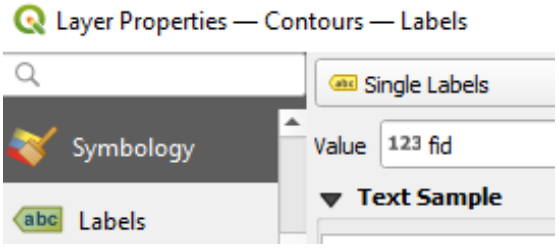
8. Right click on clipped → properties → symbology → change render type: single pseudcolor.



9. Processing → toolbox → GDAL → contour → change → run → close.



10. Right click on contour → properties → → labels → changes → apply → ok.



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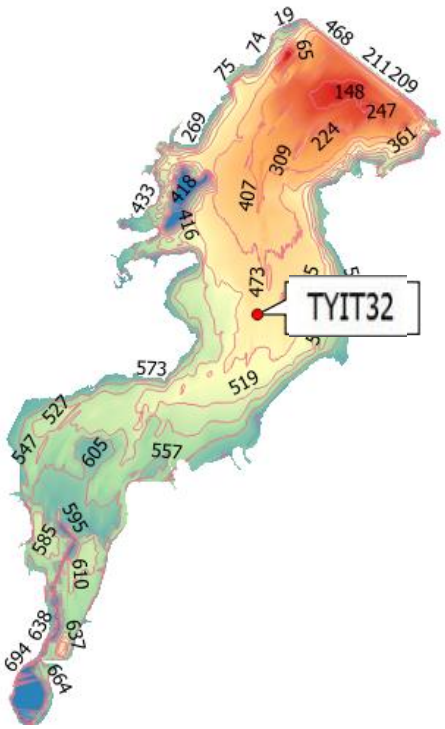
11. Output :



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